
MODEL SELECTION AND VALIDATION REPORT (only_tric_biomass model)

OBSERVATIONS:3382

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (only_tric_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	7.08	0.16
tric	0.07	0.02
sbt_mean	0.28	0.06
sic_mean	0.05	0.05
sbt_sd	0.13	0.03
sst_sd	0.06	0.03
log_chla_mean	0.03	0.03
depth	-0.03	0.03

Dispersion parameter: 0.82

Spatiotemporal AR1 correlation (rho): 0.86

Matérn range: 263.16

Spatiotemporal marginal AR1 SD: 0.84

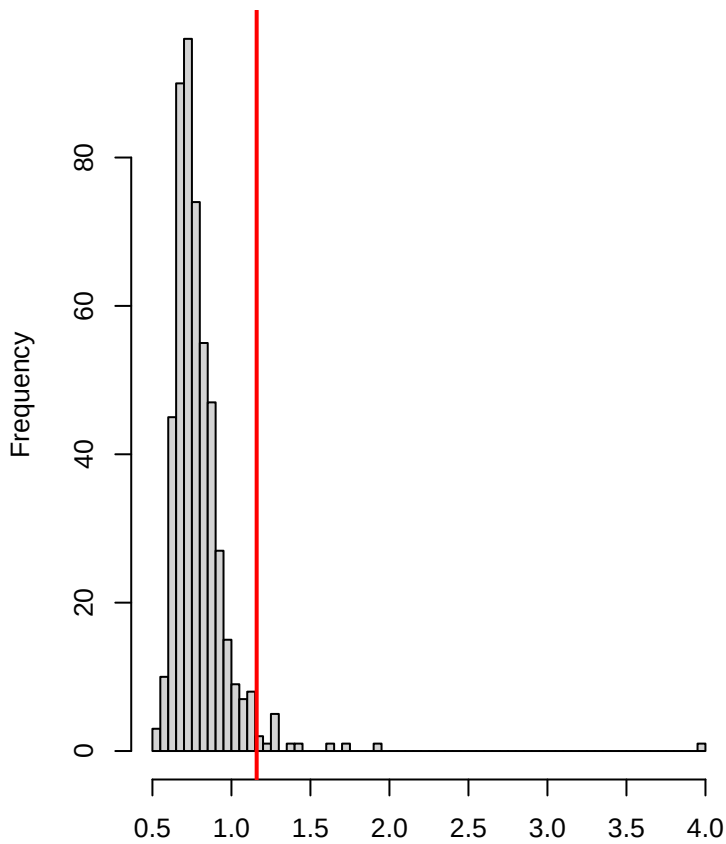
REML criterion at convergence: 27360.525

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

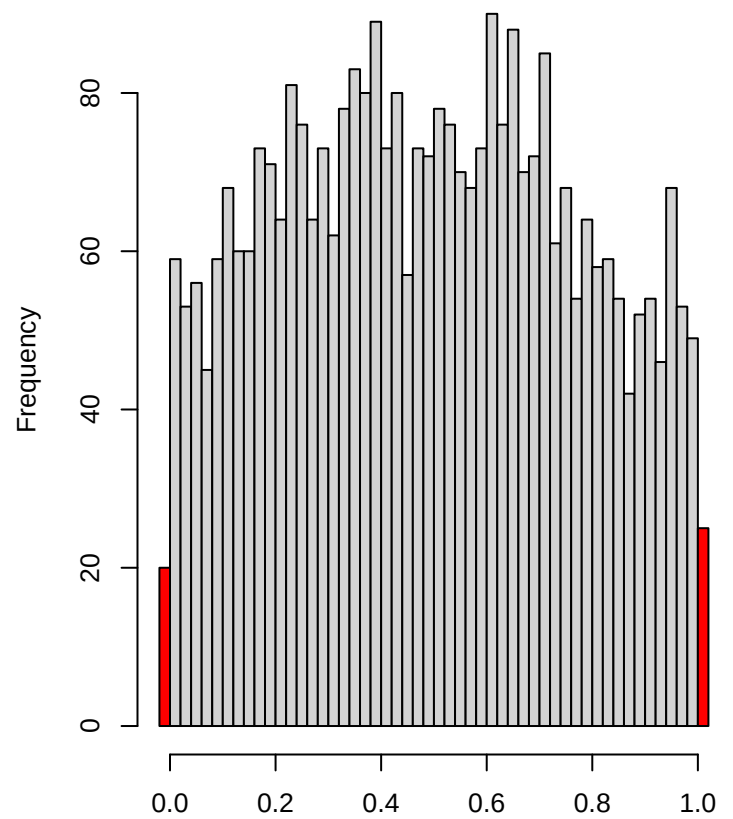
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	12	54745.05	3.318347
spatiotemporal_random_fields_ar1_with_anisotropy	14	54748.37	242.625736
spatiotemporal_random_fields_iid	11	54990.99	3.798936
spatiotemporal_random_fields_iid_with_anisotropy	13	54994.79	263.073291
spatial_random_fields	11	55257.87	2.882882
spatial_random_fields_with_anisotropy	13	55260.75	568.279484
error_only	9	55829.03	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



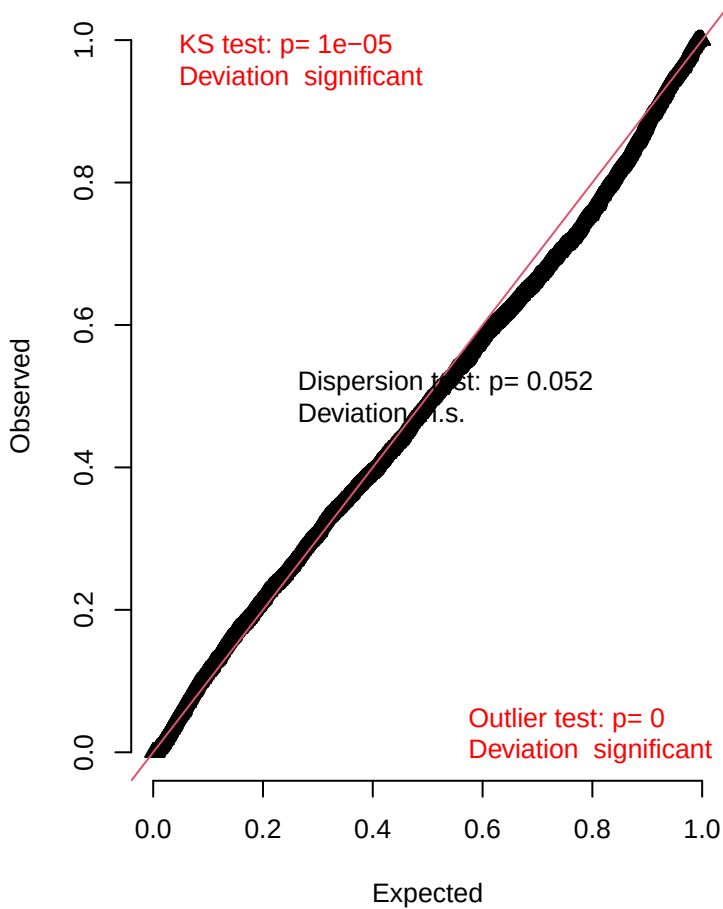
Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

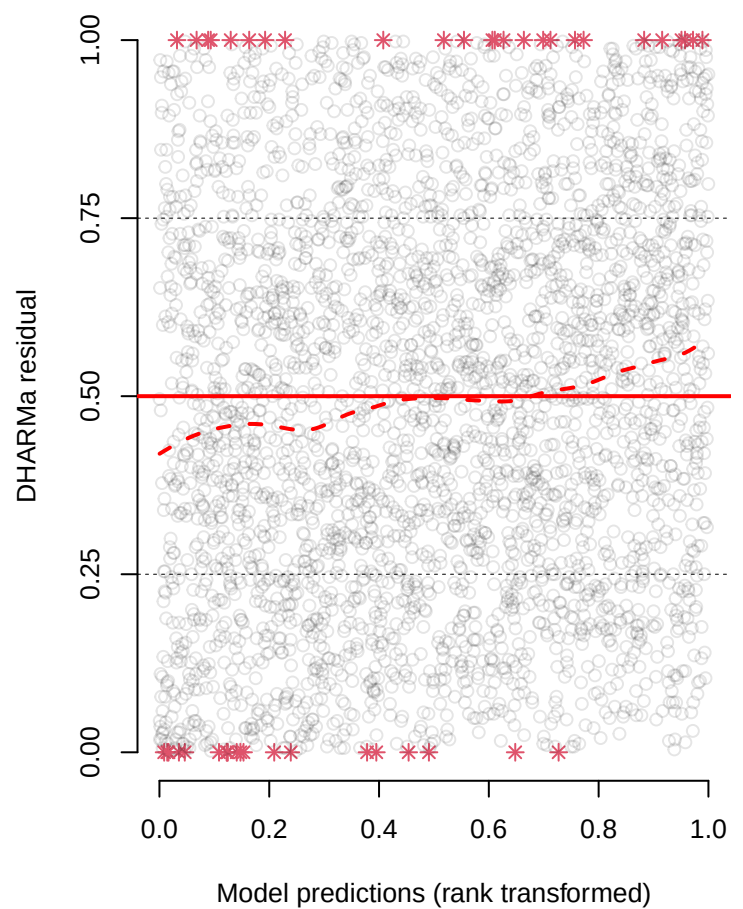


Residuals (outliers are marked red)

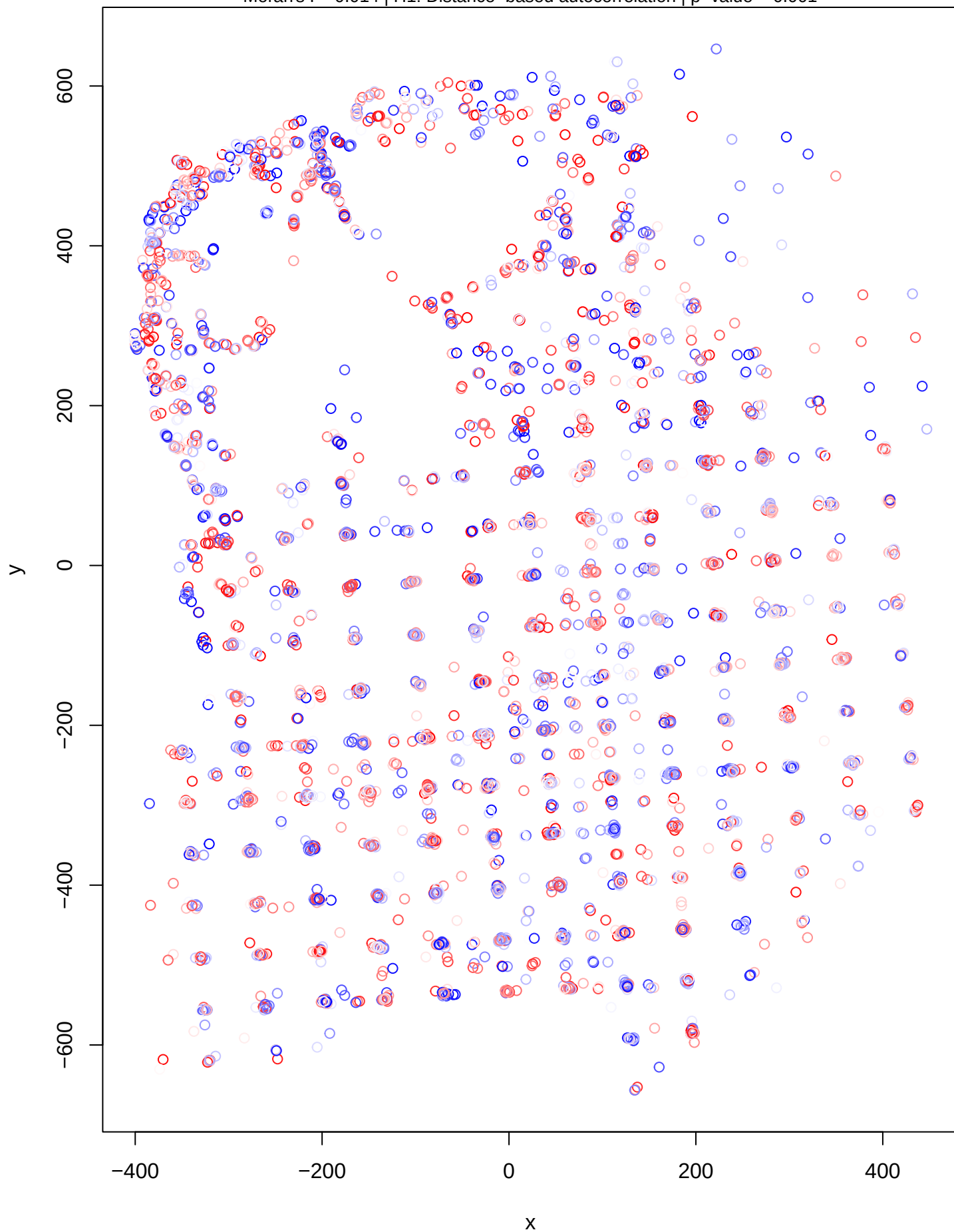
QQ plot residuals



DHARMa residual vs. predicted

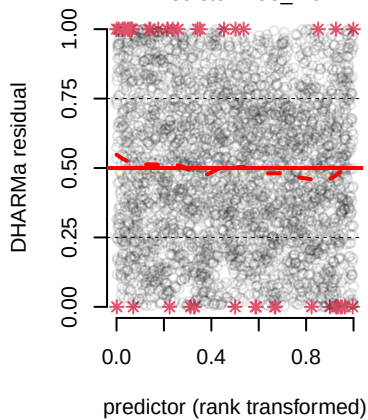


DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.014 | H1: Distance-based autocorrelation | p-value = 0.001



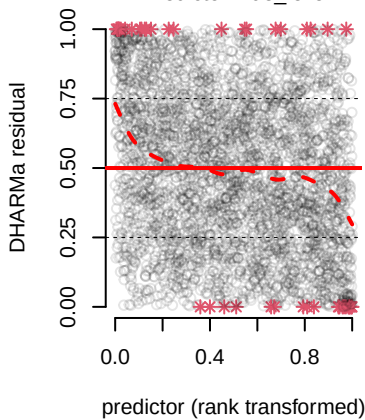
DHARMa residual Residual vs. pred

Predictor: kde_fric



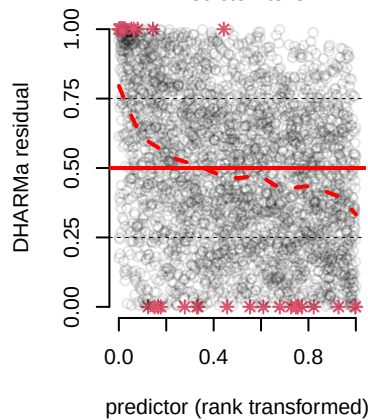
DHARMa residual Residual vs. pred

Predictor: kde_feve



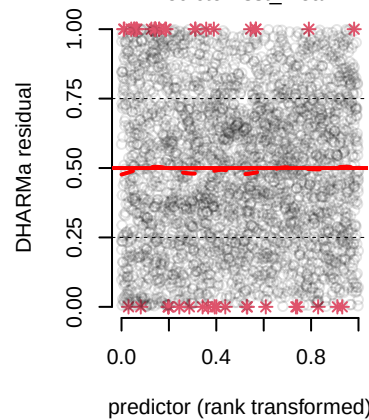
DHARMa residual Residual vs. pred

Predictor: teve



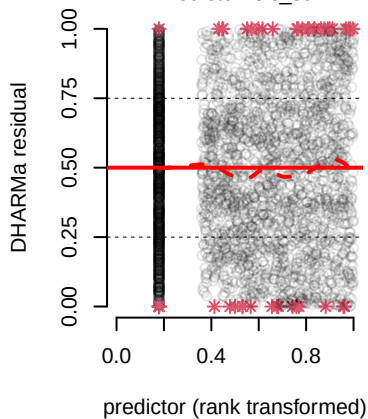
DHARMa residual Residual vs. pred

Predictor: sst_mean



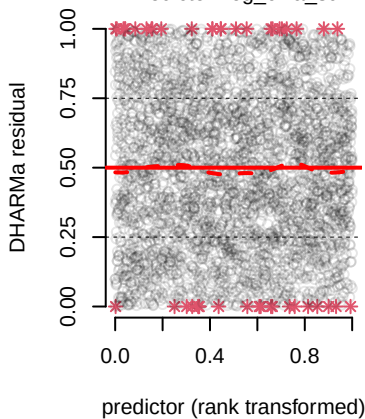
DHARMa residual Residual vs. pred

Predictor: sic_sd



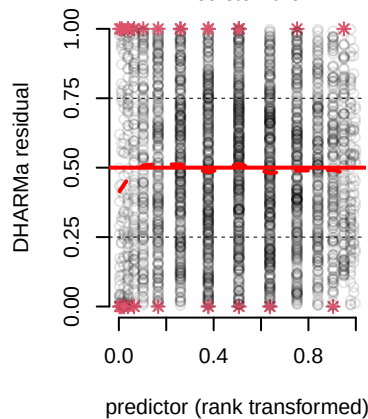
DHARMa residual Residual vs. pred

Predictor: log_chla_sd



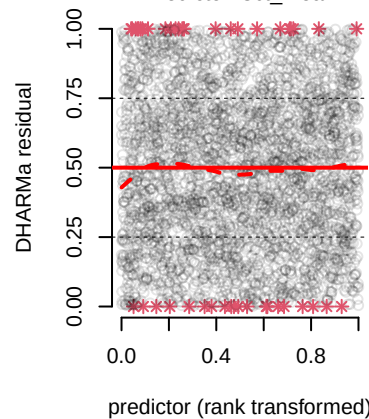
DHARMa residual Residual vs. pred

Predictor: tric



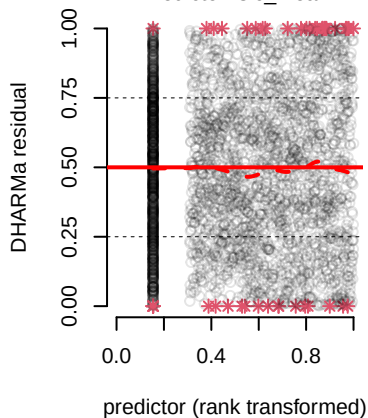
DHARMa residual Residual vs. pred

Predictor: sbt_mean



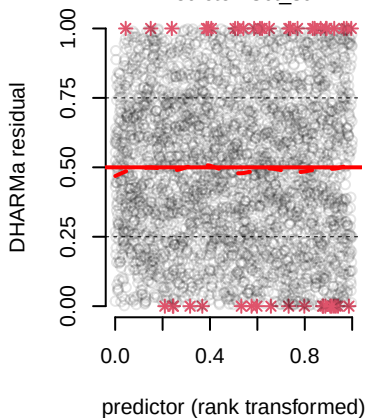
DHARMa residual Residual vs. pred

Predictor: sic_mean



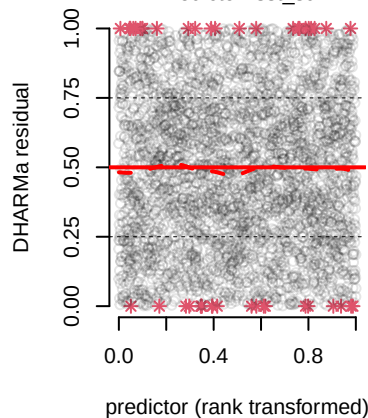
DHARMa residual Residual vs. pred

Predictor: sbt_sd



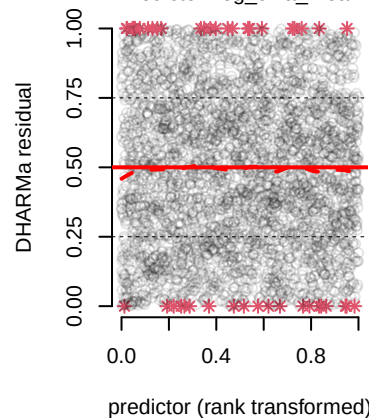
DHARMa residual Residual vs. pred

Predictor: sst_sd



DHARMa residual Residual vs. pred

Predictor: log_chla_mean



DHARMa residual Residual vs. pred

Predictor: depth

